VS-03 Data Editing and Enrichment

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VS-03 Data Editing and Enrichment

1. Purpose and value created

VS-03 is where the quality of the data is actively improved. Standardised data is rarely clean: it contains errors, inconsistencies, gaps, and values that are not yet coded. VS-03 detects these problems, corrects them, fills the gaps, and derives the additional variables that analysis needs. It is the only value stream that deliberately changes values. What makes that safe is full traceability: every change is recorded against its original value, and a reproducible, versioned dataset is produced. The value created is **data reliability and analytical readiness**. The output is Processed Data: reliable, complete microdata that estimation and modelling can build on.

VS-03 improves the quality of microdata. It does not produce estimates or aggregates: estimation, weighting, aggregation, and statistical modelling belong to VS-04 Estimation, Analysis and Statistical Modeling.

VS-03 corresponds to the business domain **Editing, derivation, and non-response correction** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

> **i** Signature discipline: full traceability
>
> Unlike VS-02, which never alters a value, VS-03 deliberately changes values to improve quality. Every edit, imputation, and derivation is rule-based, documented with a reason, attributable (who, which rule, when), and reversible. The original value is preserved beside the processed value, imputed values are flagged, and every version is reproducible. The invariant is *no undocumented or irreversible change*.

PLACEHOLDER: ArchiMate view VS-03 (a) value stream and capabilities

2. Position and state transition

VS-03 is the third primary value stream. It receives Standardized Data from VS-02 and turns it into trustworthy, analytically sound input for estimation and modelling.

VS-03 takes data from State #2 to State #3. Before VS-03, the data is structurally aligned and coded to standards, with its values unchanged. After VS-03, its content has been validated, corrected, completed, and enriched, integrated with other datasets where applicable, with new derived variables and fully documented lineage. The quality gate, the minimum metadata set, and the protection of State #3 are described in *State #3: Processed Data* and are not repeated here. Where a product skips State #2, VS-03 consumes State #1 directly.

VS-02 deliberately leaves all value-changing work to VS-03. Together with the standardised data, VS-03 receives the metadata VS-02 attached: the recorded unit, currency, and reference-period bases, the records of definition differences, and the list of unmapped or uncoded values. VS-03 then performs the operations VS-02 deferred: interpretive coding of free text and value reconciliation across sources (VS-03.2), and unit and currency conversion, reference-period alignment, and derivation (VS-03.4).

From	To	Gate	Event out
#2 Standardized Data	#3 Processed Data	Gate of State #3 (rule conformance, edit lineage, imputation flags, coding, conversion and reconciliation, linkage quality, derived variables, quality improvement, disclosure risk, reproducibility)	*Processed Data Registered*

PLACEHOLDER: ArchiMate view VS-03 (c) state handover and events: Standardized Data (#2) from VS-02, the editing and enrichment service at the domain boundary (error detection, correction, imputation, derivation, quality versioning), Processed Data (#3) handed to VS-04

3. Stages

The stages describe the value that VS-03 adds in every production cycle. Together they form the execution process type of VS-03. In a mature steady state, the rules and methods have been designed and built in advance (see attribute 4). Execution then focuses on running the cycle, applying edits, imputations, and derivations, and capturing the full edit lineage so that every value change is documented and reversible.

Stage	Name	Value added
VS-03.1	Conduct Error Detection and Quality Diagnostics	Errors, duplicates, gaps, and outliers are known
VS-03.2	Perform Data Editing and Correction	Incorrect values are corrected
VS-03.3	Impute Missing or Non-Response Data	Missing values are filled with documented methods
VS-03.4	Derive Additional Variables and Indicators	Data describes reality better than the source alone
VS-03.5	Perform Quality Evaluation and Versioning	Quality is documented, and processed microdata is registered (State #3)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-03.1 Conduct Error Detection and Quality Diagnostics

Value created: *A clear, prioritised picture of where the data is wrong, incomplete, or inconsistent.*

The stage starts when the *Standardized Data Registered* event is received from VS-02. The designed detection rules run on the incoming dataset: consistency, range, outlier, duplicate, and completeness checks. Selective-editing scores are computed, so that effort goes to the problems that matter for the results. The outcome is a diagnostic report that gives the location and type of each suspected error, its severity, and its disposition: auto-correct, flag for review, or accept. No values are changed yet. This stage diagnoses; correction happens in VS-03.2.

Activity	Key outputs	SAF process	Other reference
Run consistency, range, outlier, duplicate, and completeness checks	Error diagnostic report	PRD-5.3 Review and validate	1.3.3 Check content
Compute selective-editing scores and prioritise errors	Prioritised error list	PRD-5.3 Review and validate	
Assign a disposition: auto-correct, flag for review, or accept	Disposition list	PRD-5.3 Review and validate	
Log all detected issues to the edit audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10

VS-03.2 Perform Data Editing and Correction

Value created: *Errors corrected and inconsistencies resolved, with every change documented and reversible.*

Auto-correction rules are applied to the records that the design earmarked for automatic treatment. Flagged records go to subject-matter statisticians, who adjudicate and correct them. For every change, the editing engine records the original value, the rule or decision applied, a reason code, the actor, and a timestamp. The edit is therefore fully attributable and reversible. This stage also carries out the interpretive coding handed over from VS-02, for example coding free-text occupation to ISCO, the international occupation classification, with low-confidence cases reviewed. Where integrated sources disagree on a value, the designed value-reconciliation rules are applied. No edit may leave a record that violates the validation rules.

Activity	Key outputs	SAF process	Other reference
Apply auto-corrections according to the designed rules	Corrected records	PRD-5.4 Edit and impute	1.3.4 Adjusting and imputing
Route flagged records to subject-matter review and adjudication	Adjudicated corrections	PRD-5.4 Edit and impute	
Code free-text values interpretively (for example occupation to ISCO)	Coded values, confidence and review records	PRD-5.2 Classify and code	Handed over from VS-02
Apply value-reconciliation rules across integrated sources	Reconciled values	PRD-5.4 Edit and impute	

Activity	Key outputs	SAF process	Other reference
Record original value, rule, reason, actor, and time for every change	Edit lineage records	PRD-5.4 Edit and impute; OVA-4 Process Data Management	SAF-P10

PLACEHOLDER: VS-03 editing and correction architecture: detected issues from VS-03.1 split into three paths (auto-correctable low-impact values to automatic correction, high-impact flagged values to manual adjudication, free-text values from the VS-02 handover to interpretive coding), all producing edited values whose original value, rule, and reason are written to the edit audit trail

VS-03.3 Impute Missing or Non-Response Data

Value created: *Complete datasets where gaps are filled with the best available estimate, clearly flagged as imputed.*

Missing and non-response values are filled with the designed imputation methods. The method is chosen per variable and missingness pattern: deterministic, donor (hot-deck or nearest-neighbor), model-based, or external-source. Acceptance constraints make sure that imputed values still satisfy the edit rules. Every imputed value receives an imputation flag and a method code, so that imputed and observed values are always distinguishable downstream, and the imputation inputs are recorded in the lineage. Imputed values never overwrite observed values; they fill only genuine gaps.

Activity	Key outputs	SAF process	Other reference
Select and apply the designed imputation method per variable and missingness pattern	Imputed values	PRD-5.4 Edit and impute	1.3.4 Adjusting and imputing
Enforce acceptance constraints: imputed values satisfy the edit rules	Constraint-checked imputations	PRD-5.4 Edit and impute	
Flag every imputed value with a method code and record its inputs	Imputation flags and lineage	PRD-5.4 Edit and impute	SAF-P09
Log imputation actions to the edit audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10

VS-03.4 Derive Additional Variables and Indicators

Value created: *Enriched datasets carrying the analytical variables downstream production needs.*

The derived variables and indicators specified in design are calculated. Each one is computed from its documented formula and inputs, receives its metadata and classification, and is recorded with lineage to its inputs. This stage also computes the unit, currency, and reference-period conversions from the bases VS-02 recorded as metadata, for example converting turnover from thousands to units, or aligning a fiscal-year figure to the calendar year. Each conversion is documented with its rule and rate, so the original value remains recoverable. Where datasets are to be combined, records are linked on the designed keys, and the lineage records the linkage.

Activity	Key outputs	SAF process	Other reference
Calculate derived variables and indicators from the designed formulas	Derived variables and their metadata	PRD-5.5 Derive new variables and units	1.3.2 Deriving variables and units
Compute unit, currency, and reference-period conversions from the VS-02 bases	Converted values, conversion lineage	PRD-5.5 Derive new variables and units	
Integrate or link datasets on the designed keys, where applicable	Integrated dataset, linkage lineage	PRD-5.1 Integrate data	2.1.4 Statistical register management; SAF-P11
Record derivation and conversion formulas and inputs as lineage	Derivation lineage	PRD-5.5 Derive new variables and units; OVA-2 Metadata Management	SAF-P09

VS-03.5 Perform Quality Evaluation and Versioning

Value created: *A trustworthy, traceable Processed Data version with measured quality improvement.*

This stage evaluates the result and produces the official quality version. Quality is evaluated against the State #2 baseline: improvement in completeness and accuracy, edit and imputation rates, and the number of derived variables added. The edit lineage is verified to be complete, with every change attributable and reversible. A persistent, immutable quality version of the Processed Data is created with a unique identifier. It is registered in the catalogue with its full lineage, imputation flags, derived-variable metadata, and quality assessment, in line with SAF-P09, and access permissions are applied in line with SAF-P10. If the quality is insufficient, the data goes back for re-editing or re-imputation after review. The stage ends by publishing the *Processed Data Registered* event, which tells VS-04 that new processed data is available for estimation and analysis.

Output: Processed Data (State #3), the official handover artefact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #3: Processed Data*.

Activity	Key outputs	SAF process	Other reference
Evaluate quality improvement against the State #2 baseline	Quality evaluation report	PRD-5.8 Finalise data files; DLM-2 Promote data through a quality gate	2.1.3 Statistical quality management
Verify that the edit lineage is complete and reversible	Lineage completeness check	DLM-2 Promote data through a quality gate	SAF-P10
Create an immutable, identified quality version of Processed Data	Versioned Processed Data	PRD-5.8 Finalise data files; DLM-5 Store and version data	CSDA Traceability
Register metadata: edit lineage, imputation flags, derived-variable metadata, quality assessment	Catalogue registration	DLM-3 Register and catalogue data; OVA-2 Metadata Management	SAF-P09
Apply access permissions to the processed microdata	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10, SAF-P11
Publish the *Processed Data Registered* event to notify VS-04	Event notification	DLM-3 Register and catalogue data	SAF-P07

4. The four process types

The sub-processes below are specific to VS-03. Each refines one or more SAF business processes, listed as *Refines*, which keeps it traceable to *Business Processes*. Sub-processes marked *(standard)* work as described in Standard activities in every value stream; only the VS-03 specifics are listed.

Design: provides metadata

Without well-designed rules and methods, editing becomes ad hoc and untraceable, and management has no baseline to measure quality improvement against.

Design sub-process	Refines	Main outputs
Design Error-Detection and Validation Rules	PRD-2.5 Design processing and analysis	Validation rule set, selective-editing strategy, error taxonomy and disposition policy
Design Imputation Methodology	PRD-2.5 Design processing and analysis	Imputation methodology, donor and matching specification, imputation flagging standard
Design Derivation and Coding Specifications	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Derived-variable catalogue, coding methodology, conversion specification
Design Integration and Linkage Rules	PRD-2.5 Design processing and analysis; DLM-4 Classify and protect data	Linkage rule set, reconciliation policy, disclosure-risk assessment design
Design Quality Evaluation, Versioning, and Audit Framework	OVA-1 Quality Management; DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management	Quality criteria, edit audit-trail specification, versioning policy, access control and reproducibility design

Design Error-Detection and Validation Rules

Refines: PRD-2.5 Design processing and analysis.

The starting point of editing is defining what counts as an error. This sub-process sets the validation and error-detection rules: logical and consistency checks (between fields, within a record, and against reference data), range and outlier-detection rules, duplicate-detection rules, and completeness checks. A selective-editing strategy concentrates effort on errors that materially affect results. Score functions and thresholds decide which records are auto-corrected, which are flagged for manual review, and which are left as they are. The team also designs the error taxonomy and the disposition options: auto-correct, flag for review, or accept.

Activity	Key outputs	SAF process	Other reference
Define logical and consistency, range, and outlier-detection rules	Validation rule set	PRD-2.5 Design processing and analysis	1.3.3 Check content
Define duplicate- and completeness-detection rules	Validation rule set	PRD-2.5 Design processing and analysis	
Design selective-editing score functions and review thresholds	Selective-editing strategy	PRD-2.5 Design processing and analysis	

Activity	Key outputs	SAF process	Other reference
Define the error taxonomy and disposition options (auto-correct, review, accept)	Error taxonomy and disposition policy	PRD-2.5 Design processing and analysis	

Design Imputation Methodology

Refines: PRD-2.5 Design processing and analysis.

This sub-process designs how missing and non-response values are completed. The team selects imputation methods per variable and missingness type: deterministic or rule-based imputation, donor methods (hot-deck, nearest-neighbor), model-based methods (regression, ratio), and the use of external or administrative sources. It defines donor pools, matching variables, and acceptance constraints, such as the edit rules an imputed value must still satisfy. The design mandates an imputation flag on every imputed value, so that imputed and observed values are always distinguishable downstream.

Activity	Key outputs	SAF process	Other reference
Select imputation methods per variable and missingness type	Imputation methodology document	PRD-2.5 Design processing and analysis	1.3.4 Adjusting and imputing
Define donor pools, matching variables, and acceptance constraints	Donor and matching specification	PRD-2.5 Design processing and analysis	
Specify the use of external or administrative sources for imputation	External-source imputation rules	PRD-2.5 Design processing and analysis	
Mandate imputation flags and method codes on every imputed value	Imputation flagging standard	PRD-2.5 Design processing and analysis	SAF-P09

Design Derivation and Coding Specifications

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

This sub-process specifies the new variables and indicators to be derived, and the interpretive coding handed over from VS-02. Each derived variable is defined by a formula, its input variables, its unit, and the classification it belongs to, in line with SAF-P09. For interpretive coding, the design specifies the coding scheme (for example ISCO for occupation free text, COICOP for product names), the coding approach (manual, computer-assisted, or machine-learning-assisted with confidence thresholds), and the review workflow for low-confidence cases. This is the work VS-02 deferred because it requires judgment. The team also designs how unit, currency, and reference-period conversions are computed from the bases VS-02 recorded as metadata.

Activity	Key outputs	SAF process	Other reference
Specify derived variables and indicators: formulas, inputs, units, classification	Derived-variable catalogue	PRD-2.2 Design variables	1.3.2 Deriving variables and units
Design the interpretive coding approach and review workflow (free text to standard codes)	Coding methodology	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	

Activity	Key outputs	SAF process	Other reference
Design unit, currency, and reference-period conversion rules from the VS-02 metadata	Conversion specification	PRD-2.5 Design processing and analysis	
Define metadata and classification for every derived variable	Derived-variable metadata standard	PRD-2.2 Design variables	SAF-P09

Design Integration and Linkage Rules

Refines: PRD-2.5 Design processing and analysis; DLM-4 Classify and protect data.

Where Processed Data is to be integrated with other datasets, this sub-process designs the linkage. It specifies the linkage keys, for example the statistical register identifier aligned in VS-02, and deterministic and probabilistic matching rules. It sets conflict-resolution rules for linked sources that disagree on a value, and the value-reconciliation policy for the definition differences VS-02 recorded as metadata. Linkage can increase the disclosure risk of pseudonymised microdata, so a disclosure-risk assessment is designed in line with SAF-P11.

Activity	Key outputs	SAF process	Other reference
Define linkage keys and deterministic and probabilistic matching rules	Linkage rule set	PRD-2.5 Design processing and analysis	GSBPM 5.1
Design conflict-resolution and value-reconciliation rules across sources	Reconciliation policy	PRD-2.5 Design processing and analysis	
Design the disclosure-risk assessment for integration and enrichment	Disclosure-risk assessment design	DLM-4 Classify and protect data	SAF-P11; 3.3.1 Privacy management
Define lineage requirements for linked and reconciled values	Linkage lineage specification	PRD-2.5 Design processing and analysis	CSDA Traceability

Design Quality Evaluation, Versioning, and Audit Framework

Refines: OVA-1 Quality Management; DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management.

The last design sub-process makes sure that quality is measured and that every change is traceable. It defines the quality-evaluation criteria: improvement of completeness and accuracy against the State #2 baseline. It sets the edit audit-trail requirements: original value, rule applied, reason, actor, and timestamp for every change. The versioning policy defines how versions are identified, what triggers a new version, and immutability and retention. Role-based access control for processed microdata is designed in line with SAF-P10. Reproducibility is designed in: a quality version must be regenerable from the standardised input plus the recorded rules.

Activity	Key outputs	SAF process	Other reference
Define quality-evaluation criteria against the State #2 baseline	Quality criteria document	OVA-1 Quality Management	GAMSO Quality Management

Activity	Key outputs	SAF process	Other reference
Define edit audit-trail requirements (original value, rule, reason, actor, time)	Edit audit-trail specification	OVA-4 Process Data Management	SAF-P10
Define the versioning policy: identity, triggers, immutability, retention	Versioning policy	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control and reproducibility for processed microdata	Access control and reproducibility design	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	SAF-P10; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-03 design process decomposition: inputs (State #2 schema and metadata, methodology set for editing, imputation, and derivation, framework standards, privacy and disclosure requirements feeding the linkage design), the five design sub-processes in order, and the artefact each produces (validation rule set, imputation methodology, derivation and coding specifications, linkage and reconciliation rules, versioning and audit policy), which together govern Implementation, Execution, and Management

Implementation: provides the execution process

Implementation sub-process	Refines	Main outputs
Build Error-Detection and Editing Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Detection module, configured editing engine, edit audit-trail capture
Configure Imputation and Coding Services	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Imputation services, coding service and review queue, conversion service
Build Derivation, Integration, and Versioning Pipelines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalogue data; DLM-5 Store and version data	Derivation pipeline, integration component, versioning store, events
Test and Validate the Pipeline *(standard)*	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalise production systems	Test reports covering detectors, editing engine, imputation and coding services, derivation calculators, integration component, and versioning store; edit-impact test (edits and imputations move quality indicators in the intended direction, without over-editing); reproducibility test (a quality version is regenerated identically from the standardised input plus the recorded rules); imputation evaluated against a holdout or known-value sample

Build Error-Detection and Editing Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the validation and editing rule designs into operational tools. Error-detection components run the designed consistency, range, outlier, duplicate, and completeness checks and compute the selective-editing scores. The editing engine applies auto-corrections according to the designed rules and routes flagged records to the review workflow. For every change it captures the original value, the rule applied, and a reason code. All editing logic is version-controlled in line with SAF-P10, so that any edit is reproducible and attributable.

Activity	Key outputs	SAF process	Other reference
Build error-detection components (consistency, range, outlier, duplicate, completeness)	Production-ready detection module	PRD-3.2 Reuse or build processing and analysis components	
Build or configure the editing engine (auto-correction and review routing)	Configured editing engine	PRD-3.2 Reuse or build processing and analysis components	
Configure edit-lineage capture (original value, rule, reason, actor) for every change	Edit audit-trail capture	PRD-3.4 Configure workflows	SAF-P10
Version-control all editing logic	Versioned editing repository	PRD-3.2 Reuse or build processing and analysis components	SAF-P10

Configure Imputation and Coding Services

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

The designed imputation and coding methodologies are translated into automated services. Imputation services are configured for each method (deterministic, donor, model-based, external-source). Donor pools and matching variables are loaded, acceptance constraints are enforced, and every imputed value is flagged with its method code. The interpretive coding service handed over from VS-02 is configured: computer-assisted or machine-learning-assisted coding of free-text values with the designed confidence thresholds, and a manual review workflow for low-confidence cases. Conversion routines are configured to compute unit, currency, and reference-period conversions from the bases VS-02 recorded.

Activity	Key outputs	SAF process	Other reference
Configure imputation services per method, with donor pools and constraints	Configured imputation services	PRD-3.2 Reuse or build processing and analysis components	
Enforce imputation flagging and method codes	Flagged imputation outputs	PRD-3.4 Configure workflows	SAF-P09
Configure the interpretive coding service (free text to standard codes) with its review workflow	Coding service and review queue	PRD-3.2 Reuse or build processing and analysis components	
Configure unit, currency, and reference-period conversion routines	Conversion service	PRD-3.4 Configure workflows	

Build Derivation, Integration, and Versioning Pipelines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalogue data; DLM-5 Store and version data.

This sub-process implements derivation, dataset integration, and quality versioning. Derived-variable calculators are built from the designed formulas and produce each derived variable with its metadata and classification. The integration component is configured with the designed keys and matching rules. It applies the conflict-resolution and value-reconciliation policy and captures linkage lineage. The versioning store makes each accepted Processed Data version immutable, identified, reproducible from inputs plus rules, and accompanied by its full edit lineage. Event notifications are configured, so that VS-04 is alerted when a quality version is registered.

Activity	Key outputs	SAF process	Other reference
Build derived-variable calculators from the designed formulas	Derivation pipeline	PRD-3.2 Reuse or build processing and analysis components	
Configure integration and linkage with conflict resolution and reconciliation	Integration component	PRD-3.2 Reuse or build processing and analysis components	
Build the versioning store (immutable, identified, reproducible)	Versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Processed Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows; DLM-3 Register and catalogue data	SAF-P07

PLACEHOLDER: VS-03 implementation pipeline: the five design artefacts feeding the three build sub-processes, followed by technical testing, edit-impact and reproducibility testing, and imputation evaluation, resulting in an operational editing pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-03 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when a new standardised version is registered. The stages then run in order. Execution applies the designed rules and methods as they are; a new rule, model, or coding scheme goes back to Design. Every cycle ends in one of two outcomes: the quality version is accepted and registered, or the quality is insufficient and records go back for re-editing or re-imputation after review.

Aspect	VS-03
Trigger	*Standardized Data Registered* (or *Raw Data Registered* where State #2 is skipped)
Cadence	Per production cycle, for each new standardised version
Performed by	ROL-14 Data quality analyst, ROL-15 Subject-matter statistician, ROL-09 Classification steward, ROL-03 Data steward
Input	Standardized Data (#2) with its unmapped values and recorded units and periods; production-ready editing pipeline
Output	Registered Processed Data quality version (#3) with its edit lineage
Control point	Gate of State #3 in VS-03.5

Aspect	VS-03
Exception path	Insufficient quality: re-edit or re-impute after review; recurring problems or a need for new rules or models back to Design

Management: provides quality reports

Management sub-process	Refines	Main outputs
Monitor Process Performance *(standard)*	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard; throughput report against processing windows; paradata on rule and model versions and manual interventions
Assess Data Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Quality improvement report, imputation quality report, consistency and over-editing report, trend analysis
Manage Methodology and Version Lifecycle	OVA-2 Metadata Management; OVA-5 Knowledge Management; DLM-5 Store and version data; DLM-8 Monitor data quality, lineage and use	Methodology version register, metadata quality report, version store audit, lineage audit
Audit and Compliance Reporting *(standard)*	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Edit audit summary showing every change attributable and reversible (original value, rule, reason, actor, time); disclosure-risk report confirming that linkage on pseudonymised microdata stays within the assessed re-identification risk (SAF-P11); methodology compliance report confirming that approved editing and imputation methods were applied (4.1.5 Accountability); security compliance report on access controls and integrity of the tamper-evident audit trail
Continuous Improvement *(standard)*	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, for example refining edit rules that generate false positives, recalibrating imputation models, automating recurring manual edits, improving the coding service for free text, and tuning selective-editing thresholds to reduce over-editing; trends watched: error rates, over-editing, imputation quality, coding completion rate

Assess Data Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality improvement achieved is evaluated against the criteria set in *Design Quality Evaluation, Versioning, and Audit Framework*. Each processed dataset is assessed on

several dimensions: accuracy improvement (error rates before and after editing), completeness improvement (missingness before and after imputation), imputation quality (against holdout or known values), consistency (no residual rule violations), and coherence of derived variables. Quality indicators are produced per dataset, per source, and per cycle. Trends are tracked to detect over-editing, deteriorating sources, or imputation models that need recalibration. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate accuracy and completeness improvement against the baseline	Quality improvement report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess imputation quality against holdout or known values	Imputation quality report	OVA-1 Quality Management	CSDA Data Quality
Check for residual rule violations and over-editing	Consistency and over-editing report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across sources, variables, and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Manage Methodology and Version Lifecycle

Refines: OVA-2 Metadata Management; OVA-5 Knowledge Management; DLM-5 Store and version data; DLM-8 Monitor data quality, lineage and use.

The editing methodology and the quality versions it produces must stay current, complete, and well governed. This sub-process governs the life cycle of editing rules, imputation models, and derivation specifications: it versions them, manages transitions, and retires obsolete rules. It verifies metadata completeness as a compliance check on SAF-P09: every processed dataset must carry full edit lineage, imputation flags, and derived-variable metadata. It governs the version store: retention, immutability, and reconciliation of catalogue entries with stored versions. Metadata lineage from the standardised state through editing to the processed state is tracked. As CSDA emphasises, complete, well-managed lineage is what makes edited data trustworthy and reusable.

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of editing rules, imputation models, and derivation specifications	Methodology version register	OVA-5 Knowledge Management; OVA-2 Metadata Management	SAF-P03
Verify metadata completeness (edit lineage, imputation flags, derived-variable metadata)	Metadata quality report	OVA-2 Metadata Management	SAF-P09
Govern the version store: retention, immutability, reconciliation with the catalogue	Version store audit	DLM-5 Store and version data	CSDA Traceability
Track metadata lineage from the standardised to the processed state	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability

PLACEHOLDER: VS-03 management feedback loop: data and paradata from Execution flowing through the five management sub-processes (edit and imputation rates, quality improvement, rules and versions, edit auditability and privacy), with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, and improvement recommendations back to Design

Metadata per process type

In VS-03, metadata is what keeps value changes transparent: every edit, imputation, and derivation must be self-describing. The table shows the data metadata each process type of VS-03 contributes; the paradata are listed in Standard activities in every value stream. VS-03 also records edit-impact test outcomes in Implementation, and edit and imputation rates, rule and model versions, and manual-override records in Execution, as paradata.

Process type	Data metadata produced
Design	Editing and validation rule sets, imputation methodology, derivation formulas, integration and linkage rules, quality-version policy
Implementation	Engine configurations, rule parameters, imputation model parameters, derivation definitions, versioning templates
Execution	Per-record edit lineage (original value, rule, reason), imputation flags and methods, derived-variable formulas and inputs, quality-version identifiers
Management	Quality-improvement scores against the baseline, edit and imputation rate reports, lineage assessments, trend analyses

PLACEHOLDER: VS-03 business object lifecycle: design artefacts (validation rules, imputation methodology, derivation and coding specifications, versioning and audit policy) govern implementation assets (editing and imputation engines, derivation pipeline, integration component, versioning store), which produce the process that edits and enriches Standardized Data (#2) into Processed Data (#3), generating edit lineage, imputation flags, and quality versions that management assesses and feeds back to design

5. Actors and roles

VS-03 is mostly internal: it receives its input from VS-02 and hands its output to VS-04, and both handovers are covered by the boundary interfaces in attribute 8. The table shows the actors and the reference roles involved. Specialised role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the editing methodologist and the imputation specialist in the original VS-03 description are both ROL-07 Methodologist, and the coding specialist is ROL-09 Classification steward. The design, build, and running of the editing process are often organised in the collaboration COL-05 *Statistical product team*.

Actor or role	ID	Process type	Responsibility in VS-03
International statistical organisation	ACT-07	External	Acts as methodology authority: sets the editing, imputation, and derivation methods the NSI conforms to, for example Eurostat guidance

Actor or role	ID	Process type	Responsibility in VS-03
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-03
Data owner	ROL-02	Management	Accountable for State #3 and for the correct application of approved methods
Data steward	ROL-03	Execution	Registers processed datasets and quality versions, manages metadata entries, applies access controls
Data custodian	ROL-04	Execution	Operates storage of quality versions, access control, and retention; keeps pseudonymisation keys separate from the data
Statistical designer	ROL-06	Design	As derivation designer: specifies derived variables and indicators, formulas, and required inputs
Methodologist	ROL-07	Design	As editing methodologist: designs error-detection and editing rules, the selective-editing strategy, and quality criteria. As imputation specialist: designs imputation methodology, models, donor strategies, and imputation flags
Classification steward	ROL-09	Execution	As coding specialist: performs interpretive coding of free-text values to standard classifications, handed over from VS-02
Data engineer	ROL-10	Implementation	Builds and configures the editing, imputation, derivation, and integration engines and the pipeline
Platform engineer	ROL-11	Implementation	Configures editing services, connections to reference and donor data, and access controls
Test manager	ROL-12	Implementation	Plans and carries out testing of the editing pipeline: technical, edit-impact, and reproducibility tests

Actor or role	ID	Process type	Responsibility in VS-03
Data quality analyst	ROL-14	Execution	Runs error detection, evaluates quality improvement, and logs edit lineage and exceptions
Subject-matter statistician	ROL-15	Execution	Reviews flagged records, adjudicates edits, and signs off domain-specific corrections
Quality manager	ROL-18	Management	Monitors quality-improvement indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures edit auditability and disclosure-risk compliance and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-03 relies on, and what role they play. NSIs can use it to map the value stream to their own organisation.

Capability group	Capabilities	Role in VS-03
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the editing and imputation methodology and the derivation specifications, and build the editing pipeline
1.3 Processing	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing	Detect errors, correct and impute values, and derive new variables
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage data, edit metadata, and quality versions; assess quality improvement; integrate with statistical registers
3.2 Computerisation and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain the editing, imputation, derivation, and integration engines
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning	Manage disclosure risk in enrichment and linkage, and secure the edit audit trail
4.1 Governance	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward the editing methodology, plan improvements, report on auditability and compliance

Stages, processes, and functions

The table shows which SAF business processes realise each stage of VS-03 and which business functions they mainly use.

Stage	Realised by processes	Main functions
VS-03.1 Conduct Error Detection and Quality Diagnostics	PRD-5.3 Review and validate	FUN-3.3 Process statistical data
VS-03.2 Perform Data Editing and Correction	PRD-5.2 Classify and code; PRD-5.4 Edit and impute	FUN-3.3 Process statistical data
VS-03.3 Impute Missing or Non-Response Data	PRD-5.4 Edit and impute	FUN-3.3 Process statistical data; FUN-4.1 Manage statistical and analytical methodology
VS-03.4 Derive Additional Variables and Indicators	PRD-5.1 Integrate data; PRD-5.5 Derive new variables and units	FUN-3.3 Process statistical data
VS-03.5 Perform Quality Evaluation and Versioning	PRD-5.8 Finalise data files; DLM-2 Promote data through a quality gate; DLM-3 Register and catalogue data	FUN-3.3 Process statistical data; FUN-5.2 Assure and review quality; FUN-7.3 Manage data lifecycle and storage

PLACEHOLDER: ArchiMate view VS-03 (b) stages realised by processes and functions

Framework mappings

This reference table maps each process type of VS-03 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.2, PRD-2.5, DLM-1, DLM-4, DLM-9, OVA-1, OVA-4	Phase 2 Design (2.2, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-3, DLM-5	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Processing (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.1 to PRD-5.5, PRD-5.8, DLM-2 to DLM-5, OVA-2, OVA-4	Phase 5 Process (5.1 Integrate data, 5.2 Classify and code, 5.3 Review and validate, 5.4 Edit and impute, 5.5 Derive new variables and units, 5.8 Finalise data files)	Production (execution)	Data Processing (execution)	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing, 2.1.2 Metadata management, 2.1.3 Statistical quality management

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Management	OVA-1, OVA-2, OVA-4, OVA-5, DLM-5, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-03 most directly.

Principle	Statement	Application in VS-03
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Editing, imputation, and coding tooling, such as selective-editing engines and imputation libraries, is developed and shared in open-source communities with other NSIs. This keeps collective control over critical methodological capabilities instead of depending on vendors.
SAF-P07	Implement loosely coupled processes and systems	Editing is independent from upstream standardisation and downstream estimation. Processed Data (State #3) is the decoupling interface: downstream consumers use edited, enriched microdata without knowing how it was corrected or imputed.
SAF-P08	Data is shared property	Processed data, once versioned, is available to all authorised consumers. Editing and enrichment are not duplicated across statistical products when the same processed asset can serve several purposes.
SAF-P09	No data without metadata and classification	Every edit, imputation, and derived variable carries metadata: the rule applied, the original value, the imputation method and flag, the derivation formula, and a classification for derived variables. This makes the changes transparent and reproducible.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident edit audit trail are built into the editing pipeline from the start, so that every change to a value stays attributable and traceable.

Principle	Statement	Application in VS-03
SAF-P11	Privacy by design	VS-03 works on pseudonymised microdata. Enrichment and integration with other datasets raise the risk of re-identification, so linkage is assessed for disclosure risk and minimisation by design, not afterwards.

i Principles that apply across the process types

SAF-P09 is the backbone of the editing discipline: Design defines the rules and the required edit metadata, Execution captures the original value, rule, and reason for every change, and Management assesses edit quality and closes gaps. SAF-P10 governs the tamper-evident edit audit trail across all stages, VS-03.1 to VS-03.5. SAF-P11 applies especially to *Design Integration and Linkage Rules* and to VS-03.4, where linkage can raise the risk of re-identification. SAF-P07 governs both the intake of State #2 from VS-02 and the handover of State #3 to VS-04.

8. Interfaces, services, and events

Boundary interfaces

VS-03 has two boundary interfaces: intake from VS-02, and handover to VS-04.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-02 (or VS-01 where #2 is skipped) → VS-03 (intake)	Registered standardised data version (#2)	Source-to-target mapping lineage, applied classification versions, recorded unit, currency, and reference-period bases, list of unmapped or uncoded values for coding, records of definition differences, classification	*Standardized Data Registered*	VS-02 keeps custody of the standardised artefact; VS-03 takes responsibility for editing, imputation, derivation, interpretive coding, and value conversion on receipt of the event
VS-03 → VS-04 (handover)	Registered Processed Data quality version (#3)	Edit lineage (original value, rule, reason), imputation flags and methods, derived-variable formulas and classification, integration and linkage lineage, quality-improvement assessment, quality-version identifier, classification	*Processed Data Registered*	VS-03 keeps custody of the processed version; VS-04 takes responsibility for estimation, weighting, and analysis on receipt of the event

Business services

Service	Offered to	Realised in	Interfaces
Error-detection service	Consumers of standardised data, editing teams	VS-03.1 Conduct Error Detection and Quality Diagnostics	Diagnostic reports
Editing and correction service	Editing teams, subject-matter statisticians	VS-03.2 Perform Data Editing and Correction	Review and adjudication workbench for analysts, edit-audit query interface
Imputation service	Editing teams, methodologists	VS-03.3 Impute Missing or Non-Response Data	Imputation flags and records
Derivation and integration service	Editing teams, downstream analysts	VS-03.4 Derive Additional Variables and Indicators	Derived-variable catalogue
Processed data provisioning service	VS-04 Estimation, Analysis and Statistical Modeling, other statistical products	VS-03.5 Perform Quality Evaluation and Versioning	Event notification (*Processed Data Registered*) and a catalogue query interface to find and access processed quality versions and their edit lineage
Quality-version catalogue service	All data consumers in the NSI	Management sub-process *Manage Methodology and Version Lifecycle*	Search and browse interface for quality versions, edit lineage, imputation flags, and derived-variable metadata
Design and methodology service	Methodologists, editing and imputation designers	All Design sub-processes of VS-03	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artefacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Standardized Data Registered	Start of VS-03.1 for that version
Quality version accepted / Quality insufficient	Outcome of VS-03.5; insufficient quality starts re-editing or re-imputation after review
Processed Data Registered	Completion of VS-03.5; triggers VS-04
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-03 event choreography between the upstream value stream (VS-02), NSI teams, and process management across Design, Implementation, Execution (detect, edit, code and reconcile, impute, derive, convert and integrate, then quality version accepted or re-edit), and Management (edit and imputation rates, quality improvement, audit reports, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-03	Receives from VS-03
VS-07 Governance and Compliance	Editing and privacy policy, audit of edit lineage, methodology compliance, and disclosure risk	Edit audit summaries, methodology and disclosure-risk compliance reports
VS-08 Innovation and Methodological Development	Improved imputation and outlier-detection methods, coding tools	Improvement needs; processed data for testing new editing, imputation, or coding methods
VS-09 Infrastructure and Platform Enablement	Computing and orchestration for editing, imputation, and coding at scale; version store	Capacity and processing-window requirements

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Edit rate	Share of values changed by editing	Per variable, per cycle
Imputation rate	Share of values imputed	Per variable, per cycle
Manual-review rate	Share of records routed to manual adjudication	Per source, per cycle
Over-editing indicator	Share of edits with negligible impact on results	Per cycle
Coding completion rate	Share of free-text values successfully coded	Per classification, per cycle
Processing time	Time from standardised intake to versioned registration	Per dataset, per cycle

10. Examples

Structural Business Statistics. The design sets consistency rules (turnover 0, employees 0, turnover per employee within a plausible band by NACE class), outlier rules based on a Hidiroglou–Berthelot test on year-on-year ratios, and a selective-editing score that flags only units whose potential error would move a published aggregate by more than a set threshold. Everything else is accepted without manual work. In a cycle, the standardised SBS file is diagnosed: 312 units fail the turnover-per-employee band, 47 are duplicates, and 1,204 have missing employee counts. The selective-editing scores prioritise 89 units whose potential error would materially affect a published aggregate; the remaining low-impact flags are scheduled for automatic treatment.

Labour Force Survey imputation. For item non-response on income, about 9% of records, the design specifies regression imputation with age, education (ISCED), occupation (ISCO), and region as predictors, and a constraint that imputed income stays within the valid range of the respondent's earnings bracket. Each imputed income carries the flag `imputed=true` and the method code `REG-01`, and the predictor values used are recorded. Analysts can exclude or down-weight imputed values, and the imputation can be reproduced or revised.

Labour Force Survey coding. The 42,000 free-text occupation answers handed over from VS-02 are coded to ISCO-08. 88% are coded automatically at or above the confidence threshold of 0.85, 5,040 go to computer-assisted coding, and 1,200 ambiguous cases go to expert manual coding. Each coded value records the source text, the assigned code, the method, and the confidence, so it is auditable as an editing decision.

Price statistics. The implementation team builds an outlier detector for daily price changes, an editing engine that applies designed corrections such as decimal-point fixes, and a review queue for flagged price jumps. Every correction stores the original scraped price, the rule, and the analyst's adjudication, so the edit can be audited and reversed.

Tax register to business statistics. Turnover, recorded by VS-02 in thousands of euro on a fiscal-year basis, is converted to units on a calendar-year basis with the designed conversion and apportionment rules. The conversion factor and the original value are stored in the lineage. Turnover per employee is then derived as a new indicator, with its formula and inputs recorded.

Toolkit: SAF-TK-V03 VS-03 maturity checklist and scenarios